Technical Report: Dog Adoption Management System

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May 27, 2026

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Chapter 1

Introduction

In the contemporary digital era, the intersection of animal welfare and technology has given rise to innovative solutions aimed at addressing long-standing challenges in the pet adoption process. This technical report details the development and architecture of a sophisticated Dog Adoption Management System, a platform designed to streamline the journey from shelter to forever home.

The primary objective of this project is to create a robust, scalable, and user-centric ecosystem that empowers both animal shelters and potential adopters. By leveraging modern cloud-native technologies, microservices architecture [2], and automated CI/CD pipelines, we aim to provide a seamless experience that reduces the friction inherent in traditional adoption workflows.

Pet adoption is not merely a transactional process; it is a complex emotional and administrative journey. Shelters often struggle with fragmented data, manual record-keeping, and limited reach. Adopters, on the other hand, frequently face opaque application processes, lack of real-time updates, and difficulty in finding the right companion. This project addresses these pain points by centralizing information, automating notifications, and providing an intuitive interface for all stakeholders.

1.1 The Evolution of Animal Welfare Systems

The transition from ledger-based tracking to integrated digital ecosystems represents a paradigm shift in how animal rescues operate. Historically, the focus was purely on "shelter," providing physical safety for animals. In the modern era, the focus has shifted to "rehoming efficiency." The faster an animal can be placed in a permanent home, the more lives a shelter can save. This project is rooted in the belief that software architecture directly impacts animal welfare outcomes.

1.2 Problem Statement

Animal welfare organizations are increasingly overwhelmed by the volume of surrenders and the complexity of modern compliance and tracking requirements. Manual entry systems are prone to human error, resulting in lost medical records or outdated availability listings. Furthermore, the lack of a centralized "source of truth" leads to operational silos, where the marketing team might promote a dog that has already been adopted or is currently under medical quarantine.

1.3 The Proposed Solution

The Dog Adoption Management System proposed in this report utilizes a distributed systems approach. By breaking down the monolith into specialized services, we achieve a higher degree of fault tolerance. The use of Clean Architecture [4] ensures that the business logic remains decoupled from the specific database or cloud provider used, allowing the system to evolve over time without requiring expensive full-scale rewrites.

1.4 Future Vision and System Evolution

While the current implementation focuses on the core orchestration and service delivery, the roadmap for the Dog Adoption Management System includes two major technological leaps. First, the transition from a "3-click" rapid adoption to a rigorous, document-verified workflow ensures that every placement is backed by the necessary legal and residency documentation. Second, we envision a mobile-first expansion utilizing Computer Vision to allow users to identify dog breeds in the real world and instantly connect with available matches in our shelter network. This evolution will transform the platform from a management tool into an intelligent adoption ecosystem.

1.5 Scope and Significance

This report covers the entire lifecycle of the project, from initial case study and market analysis to technical implementation and deployment strategy. The significance of this work lies in its potential to reduce the "Length of Stay" (LOS) for animals in shelters, which is a critical metric in animal welfare. Every day an animal remains in a shelter increases the cost of care and the psychological stress on the animal [3]. By optimizing the administrative pipeline, we directly contribute to better animal outcomes and more efficient shelter operations.

Chapter 2

Case Study: Pet Adoption Challenges and Opportunities

2.1 The Landscape of Pet Abandonment

Pet abandonment remains a global crisis with profound ethical, social, and economic implications. Every year, millions of domestic animals, particularly dogs, end up in shelters or on the streets. In more developed nations, the burden falls primarily on the shelter infrastructure, whereas in developing regions, it poses a significant public health risk due to rabies and other zoonotic diseases [7].

According to global animal welfare organizations, the reasons for abandonment range from socio-economic factors to a lack of understanding of the responsibilities associated with pet ownership. In many cases, behavioral issues, life transitions, and financial constraints drive owners to surrender their companions.

2.2 The Global Crisis of Pet Overpopulation

Pet abandonment is a subset of the larger issue of pet overpopulation. Statistics indicate that in the United States alone, approximately 6.3 million companion animals enter shelters every year [1]. Of these, roughly 3.1 million are dogs. While adoption rates have been increasing, approximately 390,000 shelter dogs are still euthanized annually [1]. This is often not due to health issues, but due to a lack of space and administrative inability to process adoptions fast enough.

2.3 The Economic Cost of Abandonment

The economic impact is often overlooked. Municipalities spend billions of dollars on animal control and shelter operations. A single "intake to adoption" cycle for a dog can cost anywhere from $300 to $800, covering food, medical care, and marketing. When administrative inefficiencies delay an adoption by even a few days, the cumulative cost across hundreds of animals can be substantial.

2.4 The Psychological Dimension: Shelter Stress

Modern veterinary science has documented "shelter stress." Dogs in a shelter environment often exhibit chronic anxiety and suppressed immune systems [3]. A digital system that fast-tracks the application and matching process directly contributes to the mental well-being of the animals by reducing their "length of stay" (LOS).

2.5 Data Silos and Information Asymmetry

One of the most persistent bottlenecks identified in current shelter operations is "Information Asymmetry." Information known to the intake officer (e.g., a dog's fear of thunderstorms) often doesn't reach the marketing volunteer or the final adopter until after a placement failure occurs.

By implementing a document-based NoSQL storage for pet profiles, our system allows for the storage of unstructured, rich metadata that can be updated by different departments simultaneously. This breaks the "linear" flow of information and creates a collaborative data environment.

2.6 The Administrative Crisis in Shelters

Most animal shelters operate on very tight budgets. Staff and volunteers must manage complex intake records, medical histories, and hundreds of adoption applications. The reliance on manual or fragmented systems leads to massive inefficiencies. For example, a dog might be listed as "Available" long after an application has been approved, leading to frustration for potential adopters and redundant work for staff.

2.7 The Critical Role of Documentation in Responsible Adoption

A major challenge identified during our research is the balance between a "seamless" digital experience and the need for due diligence. While rapid adoption can reduce shelter load, a "convoluted" or rather *rigorous* documentation process is often necessary to ensure the long-term safety of the animal. This involves the verification of housing agreements, identification, and previous pet history. Future iterations of this project will integrate an automated document verification microservice to handle these complex legal requirements without increasing the manual workload of shelter staff.

2.8 Detailed Analysis of Current Workflows

In a typical non-digitized shelter, an adoption application involves a multi-step paper trail. The applicant fills out a form, which is then manually reviewed by a volunteer. This review often requires checking a physical folder for the dog's latest medical status. If the volunteer is part-time, the folder might not be accessed for 48 hours. By the time the review is complete, the dog's status may have changed. Our project aims to replace this asynchronous, error-prone workflow with a real-time, event-driven architecture that ensures all stakeholders have access to the same, current data at all times.

Chapter 3

Technical Details of Existing Solutions

To understand the value proposition of the Dog Adoption Management System, it is crucial to analyze the landscape of currently available tools. Existing solutions generally fall into three categories: Legacy Shelter Management Software, Aggregator Marketplaces, and Generic CRM/Internal tools.

3.1 Legacy Shelter Management Software

These are specialized applications designed specifically for shelters. Examples include "ShelterBuddy" and "PetPoint".

3.1.1 Architecture and Technology

Historically, these systems were built as monolithic applications. Many still rely on early cloud models that lack true elasticity. Their architectures often suffer from:

- **High Coupling:** This architectural debt makes it nearly impossible to implement rapid feature updates. A change in the intake logic might inadvertently break the adoption contract generation.
- **Data Inaccessibility:** Many legacy providers lock shelters into their ecosystem by making API access expensive or non-existent.
- **Monolithic Scaling:** If one part of the app (like image processing) is under heavy load, the entire system slows down.

3.2 Aggregator Marketplaces

Platforms like Petfinder and Adopt-a-Pet are the most visible solutions.

3.2.1 Architecture and Technology

These are modern, web-scale platforms optimized for search. They use sophisticated indexing and CDNs. However, they are primarily marketing tools, not management tools. Shelters still need a separate system to handle internal administrative data. This creates a synchronization problem where a shelter must manually update marketplaces every time a dog's status changes.

3.3 Generic CRM and Internal Tools

Some organizations use general-purpose tools like Salesforce or Trello.

3.3.1 Architecture and Technology

These tools offer flexibility through multi-tenant SaaS architectures. However, they require a "custom developer" to map pet welfare requirements to generic fields. For example, a "Lead" becomes an "Adopter." This often results in a "clunky" user experience that doesn't feel native to the mission.

3.4 Technological Stagnation in the Sector

The primary issue across all existing categories is technological stagnation. While the broader tech industry has moved toward serverless, microservices, and container orchestration [2], the animal welfare sector remains stuck with platforms that are difficult to scale and integrate. The Dog Adoption Management System leverages AKS and Docker to ensure that the infrastructure is as modern and agile as the code it runs.

3.5 Comparative Market Matrix

Feature	**Legacy SW**	**Marketplaces**	**Generic CRM**	**Our Project**
API-First	No	Partial	Yes	**Yes**
Clean Arch	No	No	N/A	**Yes**
Scalability	Low	High	High	**High**
Specialized Logic	High	Low	Low	**High**

Table 3.1: Comparison of adoption management solutions

Chapter 4

Overview of Technologies and Motivation

The Dog Adoption Management System is built using a modern, cloud-native stack designed for high availability and scalability. The selection of these technologies was driven by the need to handle complex business logic while maintaining a professional user experience.

4.1 Microservices and Clean Architecture

The project adopts a **Microservices Architecture**, which allows each functional area to be developed and scaled independently [2]. Within each service, we implement **Clean Architecture** [4]. This ensures that the core business logic is decoupled from external technical concerns like databases or web frameworks.

4.2 Backend: .NET 10

The backend is built using **.NET 10** [5]. .NET was chosen for its high performance, robust NuGet ecosystem, and first-class support for Azure integrations. We use **Entity Framework Core** for ORM and **MediatR** to implement the CQRS pattern.

4.3 Frontend: React

The UI is a Single Page Application (SPA) built with **React** and **Vite**. We use **Tailwind CSS** for a utility-first approach to styling, ensuring the platform is responsive and consistent.

4.4 Polyglot Persistence

We utilize a polyglot persistence strategy:

- **PostgreSQL:** For structured, relational data like user profiles and reviews.
- **Azure CosmosDB:** For highly scalable, document-based adoption applications [6].

4.5 Orchestration: Docker and AKS

All components are containerized using **Docker**. The system is orchestrated by **Azure Kubernetes Service (AKS)**, which provides self-healing, auto-scaling, and a managed control plane for the microservices.

4.6 DevOps and CI/CD

We use **GitHub Actions** for automated CI/CD. The pipeline is secured using **OIDC (OpenID Connect)** for password-less authentication between GitHub and Azure. On every push to the main branch, all 8 container images are built, pushed to the **Azure Container Registry**, and deployed to the cluster.

4.7 Future Technology Stack: AI and Mobile

To realize our vision for a breed-scanning mobile feature, we plan to incorporate:

- **Computer Vision (TensorFlow/PyTorch):** For implementing Convolutional Neural Networks (CNNs) that can classify dog breeds from mobile camera streams.
- **Mobile Development (React Native / Flutter):** To extend our web-based ecosystem into a cross-platform mobile application.
- **Identity and Document Verification (Azure AI Vision):** To automate the "convoluted" document verification process, ensuring that submitted IDs and housing permits are authentic and valid.

4.8 Monitoring and Observability

Azure Application Insights is integrated into the core services to provide real-time telemetry. This allows the team to monitor "Success Rates" of applications and identify performance bottlenecks in the distributed system.

Chapter 5

Business Canvas

The Dog Adoption Management System follows a classic multi-sided platform business model. Below is the Business Model Canvas detailing the strategic management and entrepreneurial elements of our platform.

- **Customer Segments**:
  - *Adopters*: Individuals or families looking to adopt a dog quickly and transparently.
  - *Animal Shelters & Rescues*: Organizations needing a digital presence and a streamlined process for managing pet adoptions.
- **Value Propositions**:
  - A streamlined, frictionless "3-click" adoption workflow.
  - Centralized and easily searchable pet listings.
  - Real-time application status tracking and notifications.
  - Transparent review system for shelters.
- **Channels**:
  - Web application (accessible via desktop and mobile devices).
  - Social media integration for sharing pet profiles.
- **Customer Relationships**:
  - Automated, self-service platform for seamless interaction.
  - Community-driven trust through a robust rating and review system.
- **Revenue Streams**:
  - Primarily a free and open-source platform.

  - Future monetization avenues could include voluntary donations, sponsorships from pet food brands, or premium SaaS features for large shelter networks.

- **Key Resources**:
  - Cloud infrastructure hosted on Microsoft Azure (AKS, SQL, Cosmos DB).
  - The software development team.
  - The underlying database of pets and shelter information.

- **Key Activities**:
  - Platform development, maintenance, and cloud orchestration.
  - Partnering with local shelters to onboard their pet inventories.
  - User support and community moderation.

- **Key Partnerships**:
  - Local animal shelters, foster networks, and veterinary clinics.
  - Cloud service providers (Microsoft Azure) for robust hosting.

- **Cost Structure**:
  - Cloud hosting and operational costs (Azure AKS, Azure Service Bus, CDN).
  - Ongoing development, testing, and system maintenance.
  - Marketing and shelter outreach operations.

Chapter 6

Architecture Diagram

The Dog Adoption Center is built using a modern, scalable microservices architecture deployed on Microsoft Azure. This approach ensures high availability, independent scaling of components, and fault isolation.

- **Frontend and Ingress**: The user interface is a web application accessible via the browser. Traffic routing and load balancing are handled by an Azure Kubernetes Service (AKS) Ingress Controller. Static assets and content delivery are accelerated using Azure CDN.
- **Microservices**: The backend logic is decoupled into several specialized APIs:
  - *UserManagementApi*: Manages user registration, authentication, and profiles. Uses Azure SQL for relational data persistence.
  - *PetManagementApi*: Handles the cataloging, searching, and filtering of pet listings. Relies on Azure SQL for structured data and Azure Blob Storage for storing pet images.
  - *AdoptionManager*: Orchestrates the core 3-click adoption workflow. Given the document-centric nature of adoption applications, this service utilizes Azure Cosmos DB for flexible, highly available storage.
  - *ReviewService*: Manages the submission and retrieval of user reviews.
  - *NotificationService*: An asynchronous worker service that listens to events via Azure Service Bus to dispatch emails and alerts to users.
  - *AnalyticsService*: Integrates with Azure Application Insights to monitor system health, gather usage telemetry, and provide analytics on adoption trends.
- **Infrastructure & DevOps**: The entire ecosystem is containerized using Docker and orchestrated via Kubernetes (AKS). Communication between services is primarily RESTful over HTTP/HTTPS, with asynchronous event-driven communication

handled by Azure Service Bus to decouple the Notification and Analytics services from the critical path.

Chapter 7

Use Case Diagram

The platform caters primarily to two types of actors: the **Adopter** (standard user) and the **Shelter Admin**. The Use Case model illustrates the boundaries of the system and the main interactions.

- **Adopter Use Cases**:
  - *Register/Login*: Create an account or authenticate into the platform.
  - *Browse and Search Pets*: View the catalog of available dogs and use filters (breed, age, size) to find a match.
  - *Submit Adoption Application*: The hallmark 3-click adoption workflow allowing users to express interest and apply for a specific dog quickly.
  - *Track Application Status*: View real-time updates on pending adoptions.
  - *Leave a Review*: Provide feedback and ratings for the shelter post-adoption.
- **Shelter Admin Use Cases**:
  - *Manage Pet Listings*: Add new dogs to the platform, upload images (stored in Azure Blob), edit details, or remove adopted pets.
  - *Process Applications*: Review incoming adoption requests from the *AdoptionManager* and approve or reject them.
  - *View Analytics*: Access dashboards powered by the *AnalyticsService* to understand adoption rates and platform engagement.

Chapter 8

Functionality, Flow, and API Documentation

The platform's functionality is exposed through a set of RESTful APIs, enabling seamless integration between the frontend and the microservices.

Core Application Flow

1. A prospective adopter visits the site and calls the *PetManagementApi* to retrieve a list of available dogs.
2. The user registers/logs in via the *UserManagementApi*, receiving an authentication token.
3. The user initiates the 3-click adoption process. The frontend submits a request to the *AdoptionManager*.
4. The *AdoptionManager* persists the application in Cosmos DB and publishes an `AdoptionRequested` event to the Azure Service Bus.
5. The *NotificationService* consumes the event and sends a confirmation email to the user and an alert to the shelter.
6. Once the shelter processes the request, the status is updated, triggering another event that notifies the user of the final decision.

Key API Endpoints

- **UserManagementApi**
  - `POST /api/users/register`: Registers a new user.
  - `POST /api/users/login`: Authenticates a user and returns a JWT.

- **PetManagementApi**
  - `GET /api/pets`: Retrieves a paginated list of pets with optional search parameters.
  - `GET /api/pets/{id}`: Retrieves detailed information and image URLs for a specific pet.
  - `POST /api/pets`: (Admin only) Adds a new pet listing and uploads images to Azure Blob Storage.
- **AdoptionManager**
  - `POST /api/adoptions/validate`: Validates identity and supporting documents without storing them.
  - `POST /api/adoptions`: Submits a new adoption application after validation.
  - `GET /api/adoptions/user/{userId}`: Retrieves all applications for a specific user.
  - `PUT /api/adoptions/{id}/status`: (Admin only) Updates the status of an application (e.g., Approved, Rejected).
- **ReviewService**
  - `POST /api/reviews`: Submits a review for a completed adoption.
  - `GET /api/reviews/shelter/{id}`: Retrieves all reviews for a specific shelter.

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